Supplementary Material

Medical Technology: A Systematic Review on Medical Devices Utilized for Epilepsy Prediction and Management

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Table S1.1. Effective Public Health Practice Project (EPHPP) Quality Assessment Tool for Quantitative Studies in Prospective Cohort studies.

Criteria		Di Lorenzo, Leyde, & Kaplan et al	Forooghifa et al	Cook et al	Kiral-Kornek et al	Nasseri et al	Weisdorf et al
Selection Bias	Q1	3	3	3	3	2	3
	Q2	5	5	1	5	5	1
Study Design	Type	3	3	2	3	3	3
Confounders	Q1	1	1	1	3	3	1
	Q2	4	4	1	4	4	1
Blinding	Q1	1	1	1	1	3	3
	Q2	3	1	1	3	3	1
Data Collection Methods	Q1	1	1	1	1	1	1
	Q2	1	1	1	1	1	1
Withdrawals and drop-outs	Q1	1	2	1	3	3	1
	Q2	2	1	1	4	4	1
Intervention Integrity	Q1	1	1	1	1	4	1
	Q2	3	1	3	1	4	1
	Q3	4	6	4	5	6	5
Analyses	Q1	4	4	4	4	4	4
	Q2	4	4	4	4	4	4
	Q3	1	1	1	1	1	1
	Q4	3	3	1	3	3	3
Final Rating		**M**	**M**	**S**	**W**	**W**	**M**

Note: S: Strong, M: Moderate, W: Weak

Table S1.2 Effective Public Health Practice Project (EPHPP) Quality Assessment Tool for Quantitative Studies in Prospective Longitudinal studies.

Criteria		**Bonnet et al**	**Gu et al**	**Johansson et al**	**Karthick, Tanaka, Khoo, & Gotman et al**	**Velez, Fisher, Bartlett, & Le et al**
Selection Bias	Q1	4	3	2	2	3
	Q2	5	1	1	1	1
Study Design	Type	3	3	3	3	3
Confounders	Q1	3	1	1	1	1
	Q2	-	4	1	1	1
Blinding	Q1	2	3	3	1	3
	Q2	1	1	1	1	1
Data Collection Methods	Q1	1	1	1	1	1
	Q2	1	1	1	1	1
Withdrawals and drop-outs	Q1	4	1	1	2	1
	Q2	4	3	3	5	1
Intervention Integrity	Q1	3	1	1	1	1
	Q2	3	1	1	1	1
	Q3	6	3	6	1	6
Analyses	Q1	4	4	4	4	4
	Q2	4	4	4	4	4
	Q3	3	1	1	1	1
	Q4	2	1	2	1	3
Final Rating		**W**	**S**	**M**	**S**	**S**

Note: Type 7 : Prospective Longitudinal, S: Strong, M: Moderate, W: Weak

Table S1.3. Effective Public Health Practice Project (EPHPP) Quality Assessment Tool for Quantitative Studies in Prospective Observational studies.

Criteria		**Sintotskiy & Hinrichs et al**	**Yamakawa et al**
Selection Bias	Q1	3	3
	Q2	4	3
Study Design	Type	8	3
Confounders	Q1	2	2
	Q2	3	4
Blinding	Q1	2	2
	Q2	1	2
Data Collection Methods	Q1	3	1
	Q2	3	1
Withdrawals and drop-outs	Q1	4	1
	Q2	5	3
Intervention Integrity	Q1	1	1
	Q2	3	1
	Q3	6	5
Analyses	Q1	4	2
	Q2	4	2
	Q3	3	1
	Q4	3	1
Final Rating		**W**	**M**

Note: Type 7: Prospective Observational, S: Strong, M: Moderate, W: Weak

Table S1.4. Effective Public Health Practice Project (EPHPP) Quality Assessment Tool for Quantitative Studies in Retrospective studies.

Criteria		**Abel et al**	**Duun-Henriksen et al**	**Arbune et al**	**Parvez & Paul**	**Sareen, Sood, & Gupta**
Selection Bias	Q1	3	3	3	3	3
	Q2	5	5	5	5	5
Study Design	Type	3	3	3	4	3
Confounders	Q1	1	1	1	1	1
	Q2	1	2	1	1	4
Blinding	Q1	3	3	3	1	1
	Q2	3	3	3	3	3
Data Collection Methods	Q1	1	1	1	1	1
	Q2	1	1	1	1	1
Withdrawals and drop-outs	Q1	2	2	2	4	4
	Q2	1	1	1	4	5
Intervention Integrity	Q1	3	1	1	4	1
	Q2	3	3	1	3	1
	Q3	4	6	5	6	5
Analyses	Q1	4	4	4	4	4
	Q2	3	3	4	4	4
	Q3	1	1	1	1	1
	Q4	1	1	1	3	3
Final Rating		**M**	**W**	**M**	**W**	**W**

Note: Type 7: Retrospective, S: Strong, M: Moderate, W: Weak

Questions

A) SELECTION BIAS

(Q1) Are the individuals selected to participate in the study likely to be representative of the target population?

1 Very likely 2 Somewhat likely 3 Not likely 4 Can't tell

(Q2) What percentage of selected individuals agreed to participate?

1 80 - 100% agreement 2 60 – 79% agreement 3 less than 60% agreement 4 Not applicable

5 Can't tell

B) STUDY DESIGN

Indicate the study design

1 Randomized controlled trial 2 Controlled clinical trial 3 Cohort analytic

4 Case-control 5 Cohort (one group pre + post) 6 Interrupted time series

7 Other specify _____ 8 Can't tell

C) CONFOUNDERS

(Q1) Were there important differences between groups prior to the intervention?

1 Yes 2 No 3 Can't tell

(Q2) If yes, indicate the percentage of relevant confounders that were controlled (either in the design (e.g. stratification, matching) or analysis)?

1 80 – 100% (most) 2 60 – 79% (some) 3 Less than 60% (few or none) 4 Can't Tell

D) BLINDING

(Q1) Was (were) the outcome assessor(s) aware of the intervention or exposure status of participants?

1 Yes 2 No 3 Can't tell

(Q2) Were the study participants aware of the research question?

1 Yes 2 No 3 Can't tell

E) DATA COLLECTION METHODS

(Q1) Were data collection tools shown to be valid?

1 Yes 2 No 3 Can't tell

(Q2) Were data collection tools shown to be reliable?

1 Yes 2 No 3 Can't tell

F) WITHDRAWALS AND DROP-OUTS

(Q1) Were withdrawals and drop-outs reported in terms of numbers and/or reasons per group?

1 Yes 2 No 3 Can't tell 4 Not Applicable

(Q2) Indicate the percentage of participants completing the study. (If the percentage differs by groups, record the lowest).

1 80 -100% 2 60 - 79% 3 less than 60% 4 Can't tell 5 Not Applicable

G) INTERVENTION INTEGRITY

(Q1) What percentage of participants received the allocated intervention or exposure of interest?

1 80 -100% 2 60 - 79% 3 less than 60% 4 Can't tell

(Q2) Was the consistency of the intervention measured?

1 Yes 2 No 3 Can't tell

(Q3) Is it likely that subjects received an unintended intervention (contamination or co-intervention) that may influence the results?

4 Yes 5 No 6 Can't tell

H) ANALYSES

(Q1) Indicate the unit of allocation

1 community 2 organization/institution 3 practice/office 4 individual

(Q2) Indicate the unit of analysis (circle one)

1 community 2 organization/institution 3 practice/office 4 individual

(Q3) Are the statistical methods appropriate for the study design?

1 Yes 2 No 3 Can't tell

(Q4) Is the analysis performed by intervention allocation status (i.e. intention to treat) rather than the actual intervention received?

1 Yes 2 No 3 Can't tell

Supplementary 2

Table S2. Cochrane Risk of Bias Tool for assessing randomized controlled/clinical trials.

	Boon et al.
Random sequence generation	U
Allocation concealment	U
Selective reporting	L
Other sources of bias	U
Blinding (participants and personnel)	H
Blinding (outcome assessment)	U
Incomplete outcome data	L
Overall	U

Note: L: low risk, H: High risk, U: unclear risk, red: high risk of bias

Table 2. Reference sheet for evaluation

Domain	Description	High risk of bias	Low risk of bias	Unclear risk of bias
Selection bias Random sequence generation	Described the method used to generate the allocation sequence in sufficient detail to allow an assessment of whether it should produce comparable groups.	Selection bias (biased allocation to interventions) due to inadequate generation of a randomized sequence.	Random sequence generation method should produce comparable groups	Not described in sufficient detail
Selection bias Allocation concealment	Described the method used to conceal the allocation sequence in sufficient detail to determine whether intervention allocations could have been foreseen in advance of, or during, enrollment	Selection bias (biased allocation to interventions) due to inadequate concealment of allocations prior to assignment.	Intervention allocations likely could not have been foreseen in advance of, or during, enrollment	Not described in sufficient detail
Reporting bias Selective reporting	Stated how the possibility of selective outcome reporting was examined by the authors and what	Reporting bias due to selective outcome reporting.	Selective outcome reporting bias not detected	Insufficient information to permit judgement (It is likely that the majority of studies will fall into this

	was found.			category.)
Other bias Other sources of bias	Any important concerns about bias not addressed above. If particular questions/entries were pre-specified in the study's protocol, responses should be provided for each question/entry	Bias due to problems not covered elsewhere in the table.	No other bias detected	There may be a risk of bias, but there is either insufficient information to assess whether an important risk of bias exists; or insufficient rationale or evidence that an identified problem will introduce bias.
Performance bias Blinding (participants and personnel)	Described all measures used, if any, to blind study participants and personnel from knowledge of which intervention a participant received. Provided any information relating to whether the intended blinding was effective.	Performance bias due to knowledge of the allocated interventions by participants and personnel during the study	Blinding was likely effective.	Not described in sufficient detail
Detection bias Blinding (outcome assessment)	Described all measures used, if any, to blind outcome assessors from knowledge of which intervention a participant received. Provided any information relating to whether the intended blinding was effective.	Detection bias due to knowledge of the allocated interventions by outcome assessors.	Blinding was likely effective	Not described in sufficient detail
Attrition bias Incomplete outcome data	Described the completeness of outcome data for each main outcome, including attrition and exclusions from the analysis.	Attrition bias due to amount, nature or handling of incomplete outcome data	Handling of incomplete outcome data was complete and unlikely to have produced bias	Insufficient reporting of attrition/exclusions to permit judgment of 'Low risk' or 'High risk' (e.g. number randomized not

	Stated whether attrition and exclusions were reported, the numbers in each intervention group (compared with total randomized participants), reasons for attrition/exclusions where reported			stated, no reasons for missing data provided)

Supplementary 3

Table S3. SYstematic Review Centre for Laboratory animal Experimentation Risk of Bias (SYRCLE RoB tool)

Studies	1	2	3	4	5	6	7	8	9	10	Overall
Coles et al	N	U	N	N	N	N	N	Y	U	U	N
Brinkmann et al	N	N	N	U	N	N	N	U	Y	U	U
Davis et al	N	U	N	U	N	N	U	Y	U	U	U
Raghunathan, Gupta, Markandeya, Roy, & Irazoqui et al	U	Y	N	U	N	Y	Y	U	Y	U	Y

Abbreviations: N, No; Y, Yes; U, unclear

Questions

1. Was the allocation sequence adequately generated and applied?

2. Were the groups similar at baseline or were they adjusted for confounders in the analysis?

3. Was the allocation adequately concealed?

4. Were the animals randomly housed during the experiment?

5. Were the caregivers and/or investigators blinded from knowledge which intervention each animal received during the experiment?

6. Were animals selected at random for outcome assessment?

7. Was the outcome assessor blinded?

8. Were incomplete outcome data adequately addressed?

9. Are reports of the study free of selective outcome reporting?

10. Was the study apparently free of other problems that could result in high risk of bias?